

Haogao GU

722 W 168th St,
New York, NY 10032

E-mail: work@guhaogao.com
Website: guhaogao.com

EDUCATION

- **PhD (Infectious Disease), The University of Hong Kong** Sep. 2016 – Oct. 2020
- **BMed (Preventive Medicine), Sun Yat-sen University** Sep. 2011 – Jun. 2016

EXPERIENCE

- **Associate Research Scientist (Mailman School of Public Health), Columbia University** Jun. 2026 – Present
PI: Sen Pei, Department of Environmental Health Sciences
- **Research Assistant Professor (School of Public Health), The University of Hong Kong** Jul. 2024 – Jun. 2026
- **Visiting Scholar (Evolutionary dynamics of infectious diseases), Institut Pasteur** Dec. 2024 – Feb. 2025
Host: Sebastian Duchene, Department of Computational Biology
- **Postdoctoral Fellow (School of Public Health), The University of Hong Kong** Jan. 2021 – Jun. 2024
Mentor: Leo Poon, Department of Laboratory Sciences

HONORS AND AWARDS

- **Gold Medal, 51st International Exhibition of Inventions at Geneva** Mar. 2026
Core member. Project: A broad-spectrum live-attenuated mucosal vaccine against beta-coronaviruses
- **The HKU-Pasteur Fellowship (4th edition), HKU-Pasteur Research Pole** Apr. 2024
- **Faculty Outstanding Research Output Award, The University of Hong Kong** Feb. 2024
awarded twice, for two separate projects
- **Outstanding Project Team, on COVID-19 research award, Food and Health Bureau (Hong Kong)** Nov. 2021
- **Postgraduate Scholarship, The University of Hong Kong** 2016–2020
- **Best Student Poster, International Symposium for One Health Research (Guangzhou)** 2014
- **Individual Scholarship, Sun Yat-sen University** 2013
- **The Third Prize Scholarship, Sun Yat-sen University** 2012
- **First Prize, Chinese National Olympiad in Biology** Sep. 2010
qualified for guaranteed admission to Sun Yat-sen University

GRANTS

- **General Research Fund (GRF), Research Grants Council, Hong Kong SAR** 2026–2027
Early Prediction of SARS-CoV-2 Variant Dynamics: Modelling Virus Interactions with Antibodies and ACE2 Receptor.
Role: **Co-I** (PI: Hsiang-Yu Yuan, City University of Hong Kong) \$840,217 (HKD)
- **RGC Postdoctoral Fellowship Scheme, University Grants Committee, Hong Kong SAR** 2023–2026
Quantifying attenuation effects caused by dinucleotide biases and codon pair biases for development of safe and scalable live attenuated virus vaccines.
Role: **PI** \$1,258,250 (HKD)
- **CEIRR Option (University of Georgia), NIH, United States** 2023–2025
Investigation of repeated influenza vaccination for epitope focusing.
Role: **Key personnel** (PI: Sophie Valkenburg) \$454,428 (USD)

TEACHING

- **Postgraduate Certificate in Bioinformatics for MLT (LMLS6010), HKU SPACE** Jan. 2026
Lecturer for 12 classes on Programming for Biological Data
- **Scientific Skills (LMLS6003), MSc in Biomedical Science, HKU SPACE** Oct. 2024
Lecturer for 6 classes on Data Handling in Bioinformatics
- **Applied Medical Microbiology (LMLS6004), MSc in Biomedical Science, HKU SPACE** Nov. 2024
Lecturer for 2 classes on Bioinformatic Tutorial

PUBLICATIONS

Summary

Papers in peer-reviewed journals: **46** (citations: **2566**; h-index: **19**)
- first-/last- authored papers **16**

Journal papers (in reverse chronological order):

1. **Gu H**, Li J, Sun W, Li M, Leung K, Wu JT, Yuan HY, Wang MH, Yang B, McKay MR, Ning N & Poon LLM. *Optimizing global genomic surveillance for early detection of emerging SARS-CoV-2 variants*. **Nature Communications**. (2026)
2. Song ZY, Zhang J, Shum MHH, **Gu H**, Shi W, Siu GKH, Poon LLM, Wu JT, Leung KSM & Lam TTY. *Geographical dispersion and transmission of multiple SARS-CoV-2 variants in Hong Kong*. **Urban Informatics**, 5(1), 20. (2026)
3. **Gu H**, Li M, Wang R, Lee LK, Tam WY, Ho AYM, Yau MCY, Yip KT, Wong RCW, Wong BKC, Luk KS, Lam JYW, Que TL, Chow VCY, Chau SKY, Duchene S, Siu GKH & Poon LLM. *Genomic Epidemiology of Hospital-Associated SARS-CoV-2 Clusters in Hong Kong During a Period of Relaxed Visitation (May–August 2022)*. **Influenza and Other Respiratory Viruses**, 20(3), e70249. (2026)
4. Ye Y, Shuai H, Song Y, Shum MHH, Smith DK, **Gu H**, Zhu H, Wu JT, Lewis NS, Bonfante F, Holmes EC, Peiris M, Poon LLM, Guan Y & Lam TTY. *Genomic features associated with sustained mammalian transmission of avian influenza A viruses*. **Nature Microbiology**, s41564-025-02257-4. (2026)
5. Te N, Chin AWH, **Gu H**, Jia JZ, Huang Y, Cheng SMS, Tan CW, Zhao J, Hui KPY, Chan MCW, Feng B, Quadeer AA, McKay MR, Wang L, Valkenburg SA, Peiris M & Poon LLM. *Broad beta-CoV immunity and transmission blockade by a single-dose live-attenuated vaccine with atypical codon usage*. **Proceedings of the National Academy of Sciences**, 123(4), e2518645123. (2026)
6. Tuohetesen S, Qu Y, Hopke PK, Zhang K, Liu Y, Lin S, **Gu H**, Wang X, Lau SSS, Lin X, Gao X, Wu Y, Zhou X, Lin Z, Zhang M, Sun Y, Liu X, Chen J & Zhang W. *Potential impact of maternal nighttime light exposure and its interaction with sociodemographic characteristics on the risk of various congenital heart diseases*. **Epidemiology**, 36(5), 625–635. (2025)
7. Zhang Y, Wei J, Chen S, Benmarhnia T, Zhang K, Wang X, Deng X, **Gu H**, Lin Z, Qu Y, Xiao J, Jiang J, Du Z, Zhang W & Hao Y. *Individual and mixed associations between fine particulate matter components and hospital admissions for hypertension: Insights from a large-scale South Chinese cohort study*. **Sustainable Cities and Society**, 106293. (2025)
8. Wu G, Wang S, Wu W, Benmarhnia T, Lin S, Zhang K, Xue Romeiko X, **Gu H**, Qu Y, Xiao J, Deng X, Lin Z, Du Z, Zhang W & Hao Y. *Potential causal links of long-term PM2.5 components exposure with diabetes incidence and mortality in the United States*. **Sustainable Cities and Society**, 106071. (2025)
9. Sun W, Choy KT, Cheng KM, Brackman CJ, Cheng SMS, Sit THC, Tse ACN, Sims LD, **Gu H**, Tang AWY, Wong ANC, Tsang ATL, Koo JCT, Luk LLH, Yen HL, Peiris M & Poon LLM. *Detection and characterisation of high pathogenicity avian influenza virus (H5N1/H5N8) clade 2.3. 4.4 b, Hong Kong SAR, China, 2021 to 2024*. **Eurosurveillance**, 2400839. (2025)
10. Cao Y, Chin AWH, **Gu H**, Li M, Gu Y, Lau SPN, Hui KPY, Chan MCW & Poon LLM. *An interferon-stimulated long non-coding RNA USP30-AS1 as an immune modulator in influenza A virus infection*. **PLOS pathogens**, e1012854. (2025)
11. Wang X, Zhang Y, Benmarhnia T, Zhang K, Lin S, Deng X, **Gu H**, Wu W, Lin X, Du Z, Wang Y, Qu Y, Lin Z, Zhang M, Sun Y, Dong GH, Wei Y, Zhang W & Hao Y. *The effect of residential greenness on cardiovascular mortality from a large cohort in South China: An in-depth analysis of effect modification by multiple demographic and lifestyle characteristics*. **Environment International**, 108894. (2024)
12. Nelson C, Poon LLM & **Gu H**. *Reply to: Population genetic considerations regarding the interpretation of within-patient SARS-CoV-2 polymorphism data*. **Nature Communications**, 15, 3239. (2024)
13. Jia J*, Cohen C*, **Gu H***, McLean M, Varadarajan R, Bhandari N, Peiris M, Leung G, Poon L, Chung A, Cowling B, Leung N, Tsang T & Valkenburg S. *Influenza antibody breadth and effector functions are immune correlates from acquisition of pandemic infection of children*. **Nature Communications**, 15, 3210. (2024)
14. Cheng SMS, Mok CKP, Li JKC, Chan KKP, Luk KS, Lee BHW, **Gu H**, Chan KCK, Tsang LCH, Yiu KYS, Ling KKC, Tang YS, Luk LLH, Yu JKM, Pekosz A, Webby RJ, Cowling BJ, Hui DSC & Peiris M. *Cross-neutralizing antibody against emerging Omicron subvariants of SARS-CoV-2 in infection-naïve individuals with homologous BNT162b2 or BNT162b2 (WT+ BA. 4/5) bivalent booster vaccination*. **Virology Journal**, 21(1), 1–7. (2024)
15. Xu X, Deng Y, Ding J, Shi X, Zheng X, Wang D, Yang Y, Liu L, Wang C, Li S, **Gu H**, Poon LLM & Zhang T. *Refining detection methods for emerging SARS-CoV-2 mutants in wastewater: A case study on the Omicron variants*. **Science of The Total Environment**, 904, 166215. (2023)
16. Xu X, Deng Y, Ding J, Zheng X, Wang C, Wang D, Liu L, **Gu H**, Peiris M, Poon LLM & Zhang T. *Wastewater genomic sequencing for SARS-CoV-2 variants surveillance in wastewater-based epidemiology applications*. **Water Research**, 244, 120444. (2023)
17. Zheng MZM, Tan TK, Villalon-Letelier F, Lau H, Deng YM, Fritzlar S, Valkenburg SA, **Gu H**, Poon LLM, Reading PC, Townsend AR & Wakim LM. *Single-cycle influenza virus vaccine generates lung CD8+ Trm that cross react against viral variants and subvert virus escape mutants*. **Science Advances**, 9 (36), eadg3469. (2023)
18. Xie R, Edwards KM, Adam DC, Leung KSM, Tsang TK, Gurung S, Xiong W, Wei X, Ng DYM, Liu GYZ, Krishnan P, Chang LDJ, Cheng SMS, **Gu H**, Siu GKH, Wu JT, Leung GM, Peiris M, Cowling BJ, Poon LLM & Dhanasekaran V. *Resurgence of Omicron BA.2 in SARS-CoV-2 infection-naïve Hong Kong*. **Nature Communications**, 14, 2422. (2023)
19. **Gu H**, Quadeer AA, Krishnan P, Chang L, Liu G, Ng D, Cheng S, Lam T, Peiris M, McKay M & Poon L. *Within-host*

- genetic diversity of SARS-CoV-2 lineages in unvaccinated and vaccinated individuals. **Nature Communications**, 14, 1793. (2023)
20. Adam D, Martín-Sánchez M, **Gu H**, Wu P, Lau E, Leung G, Poon L & Cowling B. Risk of within-hotel transmission of SARS-CoV-2 during on-arrival quarantine in Hong Kong: an epidemiological and phylogenomic investigation. **The Lancet Regional Health - Western Pacific**, 100678. (2023)
 21. Chu JTS, **Gu H**, Sun W, Fan RLY, Nicholls JM, Valkenburg SA & Poon L. Heterosubtypic immune pressure accelerates emergence of influenza A virus escape phenotypes in mice. **Virus Research**, 198991. (2022)
 22. Leong N, **Gu H**, Ng YM, Chang L, Krishnan P, Cheng S, Peiris M & Poon L. Development of multiplex RT-ddPCR assays for detection of SARS-CoV-2 and other common respiratory virus infections. **Influenza and Other Respiratory Viruses**, 1 (11), 13084. (2022)
 23. Jia J, Tan CW, Cheng S, **Gu H**, Yeoh A, Mok CKP, Wang Y, Zhao J, Leung N, Cowling B, Poon L, Hui D, Wang LF, Peiris M & Valkenburg S. Priming conditions shape breadth of neutralizing antibody responses to sarbecoviruses. **Nature Communications**, 13, 6285. (2022)
 24. Hui K, Ng KC, Ho JCW, Yeung HW, Ching RHH, **Gu H**, Chung JCK, Chow VLY, Sit KY, Hsin MKY, Au TWK, Poon L, Peiris JS, Nicholls J & Chan M. Replication of SARS-CoV-2 Omicron BA. 2 Variant in Ex Vivo Cultures of the Human Upper and Lower Respiratory Tract. **eBioMedicine**, vol. 83: 104232. (2022)
 25. Su W, Choy KT, **Gu H**, Sia SF, Cheng KM, Nizami SIN, Krishnan P, Ng YM, Chang L, Liu Y, Cheng S, Peiris M, Poon L, Nicholls J & Yen HL. Reduced Pathogenicity and Transmission Potential of Omicron BA.1 and BA.2 Sublineages Compared with the Early Severe Acute Respiratory Syndrome Coronavirus 2 D614G Variant in Syrian Hamsters. **Journal of Infectious Diseases**, jiac276. (2022)
 26. Sun W, Cheng S, Lam K, Kwan T, Wong R, Lau L, Liu G, Luk L, Li J, **Gu H**, Peiris M & Poon L. Natural Reassortment of Eurasian Avian-Like Swine H1N1 and Avian H9N2 Influenza Viruses in Pigs, China. **Emerging Infectious Diseases**, 28.7: 1509. (2022)
 27. Bull MB, **Gu H**, Ma F, Perera L, Poon L & Valkenburg S. Next Generation T cell-activating vaccination increases influenza virus mutation prevalence. **Science Advances**, 8(14), eabl5209. (2022)
 28. Hachim A, **Gu H**, Kavian O, Mori M, Kwan M, Chan WH, Yau YS, Chiu S, Tsang O, Hui D, Mok C, Ma F, Lau E, Amarasinghe G, Qavi A, Cheng S, Poon L, Peiris M, Valkenburg S & Kavian N. SARS-CoV-2 accessory proteins reveal distinct serological signatures in children. **Nature Communications**, 13: 2951. (2022)
 29. Yen HL, Sit T, Brackman C, Chuk S, **Gu H**, Tam K, Law P, Leung G, Peiris M, Poon L, Cheng S, Chang L, Krishnan P, Ng D, Liu G, Hui M, Ho SY, Su W, Sia SF, Choy KT, Cheuk S, Lau S, Tang A, Koo J & Yung L. Transmission of SARS-CoV-2 delta variant (AY. 127) from pet hamsters to humans, leading to onward human-to-human transmission: a case study. **The Lancet**, 399.10329: 1070–1078. (2022)
 30. **Gu H***, Xie R*, Adam D*, Tsui J*, Chu D*, Chang L, Cheuk S, Gurung S, Krishnan P, Ng D, Liu G, Wan C, Cheng S, Edwards K, Leung K, Wu J, Tsang D, Leung G, Cowling B, Peiris M, Lam T, Dhanasekaran V & Poon L. Genomic epidemiology of SARS-CoV-2 under an elimination strategy in Hong Kong. **Nature Communications**, 13, 736 : 1–10. (2022)
 31. Hui K, Ho J, Cheung MC, Ng KC, Ching R, Lai KL, Kam T, **Gu H**, Sit KY, Hsin M, Au T, Poon L, Peiris M, Nicholls J & Chan M. SARS-CoV-2 Omicron variant replication in human respiratory tract ex vivo. **Nature**, 603: 715–720. (2022)
 32. **Gu H**, Ng D, Liu G, Cheng S, Krishnan P, Chang L, Cheuk S, Hui M, Lam T, Peiris M & Poon L. Recombinant BA. 1/BA. 2 SARS-CoV-2 Virus in Arriving Travelers, Hong Kong, February 2022. **Emerging infectious diseases**, 28 (6) : 1276. (2022)
 33. **Gu H**, Krishnan P, Ng D, Chang L, Liu G, Cheng S, Hui M, Fan M, Wan J, Lau L, Cowling B, Peiris M & Poon L. Probable Transmission of SARS-CoV-2 Omicron Variant in Quarantine Hotel, Hong Kong, China, November 2021. **Emerging infectious diseases**, 28 (2): 460. (2022)
 34. **Gu H**, Cheng S, Krishnan P, Ng D, Chang L, Liu G, Cheuk S, Hui M, Fan M, Wan J, Lau L, Chu D, Dhanasekaran V, Peiris M & Poon L. Monitoring International Travelers Arriving in Hong Kong for Genomic Surveillance of SARS-CoV-2. **Emerging infectious diseases**, 28(1): 247–250. (2022)
 35. **Gu H**, Chu D, Chang L, Cheuk S, Gurung S, Krishnan P, Ng D, Liu G, Wan C, Xie R, Cheng S, Cowling B, Tsang D, Peiris M, Dhanasekaran V & Poon L. Genetic Diversity of SARS-CoV-2 among Travelers Arriving in Hong Kong. **Emerging infectious diseases**, 27(10): 2666–2668. (2021)
 36. Dhanasekaran V, Edwards K, Xie R, **Gu H**, Adam DC, Chang L, Cheuk S, Gurung S, Krishnan P, Ng D, Liu G, Wan C, Cheng S, Tsang D, Cowling B, Peiris M & Poon L. Air travel-related outbreak of multiple SARS-CoV-2 variants. **Journal of Travel Medicine**, taab149. (2021)
 37. Chu D*, **Gu H***, Chang L, Cheuk S, Gurung S, Krishnan P, Ng D, Liu G, Wan C, Tsang D, Peiris M & Poon L. SARS-CoV-2 superspread in fitness center, Hong Kong, China, March 2021. **Emerging infectious diseases**, 27(8): 2230–2232. (2021)

38. Chu D*, Hui K*, **Gu H**, Ko RLW, Krishnan P, Ng D, Liu G, Wan C, Cheung MC, Ng KC, Nicholls J, Tsang D, Peiris M, Chan M & Poon L. *Introduction of ORF3a-Q57H SARS-CoV-2 variant causing fourth epidemic wave of COVID-19, Hong Kong, China. Emerging infectious diseases*, 27(5): 1492–1495. (2021)
39. **Gu H**, Chu D, Peiris M & Poon L. *Multivariate analyses of codon usage of SARS-CoV-2 and other betacoronaviruses. Virus Evolution* 6, no. 1: veaa032. (2020)
40. Sah R, Rodriguez-Morales AJ, Jha R, Chu D, **Gu H**, Peiris M, Bastola A, Lal BK, Ojha HC, Rabaan AA, Zambrano LI, Costello A, Morita K, Pandey BD & Poon L. *Complete genome sequence of a 2019 novel coronavirus (SARS-CoV-2) strain isolated in Nepal. Microbiology Resource Announcements* 9, no. 11. (2020)
41. **Gu H**, Fan R, Wang D & Poon L. *Dinucleotide evolutionary dynamics in influenza A virus. Virus Evolution* 5, no. 2: 2019 vez038. (2019)
42. Al-Ghadeer H, Chu D, Rihan EMA, Abd-Allah EM, **Gu H**, Chin A, Qasim IA, Aldoweriej A, Alharbi SS, Al-Aqil 2018 MA, Al-Sahaf A, Abdel Rahman SS, Aljasssem AH, Abdul-Al A, Aljasir MR, Alhammad YMO, Kasem S, Peiris M, Zaki AZSA & Poon L. *Circulation of Influenza A (H5N8) Virus, Saudi Arabia. Emerging Infectious Diseases* 24, no. 10: 1961. (2018)
43. **Gu H**, Leung RKK, Jing Q, Zhang W, Yang Z, Lu J, Hao Y & Zhang D. *Meteorological Factors for Precise Dengue Fever Control and Prevention in South China. International Journal of Environmental Research and Public Health*, 13(9), 867. (2016)
44. Chen T*, **Gu H***, Leung RKK, Liu R, Chen Q, Wu Y & Li Y. *Evidence-Based interventions of Norovirus outbreaks in China. BMC Public Health*, 16(1): 1072. (2016)
45. **Gu H**, Zhang W, Xu H, Li P, Wu L, Guo P, Hao Y, Lu J & Zhang D. *Predicating risk area of human infection with avian influenza a (H7N9) virus by using early warning model in China. Chinese Journal of Epidemiology*, 36(5): 470–475. (2015)
46. Li P, Xu H, **Gu H**, Ma M, Lu J, Hao Y & Zhang D. *Epidemiological Characteristics of Influenza Virus H7N9 and Evolutionary Analysis of Hemagglutinin and Neuraminidase Genes. Journal of Sun Yat-sen University (Medical Sciences)*, 06. (2014)

SUPERVISION

Co-supervision:

- Ruixuan Wang, PhD student, The University of Hong Kong Feb. 2025 – May 2026
Within-host Viral Phylodynamics
- Mengting Li, PhD student, The University of Hong Kong Oct. 2025 – May 2026
Virtual Cell and Viral Infection

INVITED TALKS

- **Genomic Surveillance and Molecular Epidemiology of SARS-CoV-2 in Hong Kong.** Jul. 2025
XLII Annual Meeting of the Willi Hennig Society, The Chinese University of Hong Kong, HKSAR, China

PROFESSIONAL DEVELOPMENT

Workshops:

- *Workshop on Molecular Evolution*, Marine Biological Laboratory (MBL), Woods Hole, MA, USA May 2025
- *Virus Evolution and Epidemiology Summer Course*, Croucher Foundation, Hong Kong Jul. 2024
- *Models and Inference in Population Genetics IV*, University of Warwick, UK Apr. 2024

Online specializations:

- *Discrete Mathematics for Computer Science*, University of California San Diego Oct. 2019
- *Bayesian Statistics and Modelling*, University of California Santa Cruz Aug. 2019
- *Python 3 Programming*, University of Michigan Jun. 2019
- *Mathematics for Machine Learning*, Imperial College London Apr. 2019

Teaching aptitude:

- Certificate of Teaching and Learning in Higher Education, CETL, HKU Dec. 2024
- PBL Tutor Training Certificate, LKS Faculty of Medicine, The University of Hong Kong Oct. 2025

ACADEMIC SERVICES

Managing editor:

Virology Journal

Ad hoc peer reviewer:

SCIENCE
COMMUNICATIONS MEDICINE
PLOS PATHOGENS
FRONTIERS IN IMMUNOLOGY
VIRUS EVOLUTION
NPJ VIRUSES
VIROLOGY JOURNAL
BMC MEDICAL RESEARCH METHODOLOGY
JOURNAL OF MULTIDISCIPLINARY HEALTHCARE

THE LANCET MICROBE
EMERGING INFECTIOUS DISEASES
CELLULAR AND MOLECULAR LIFE SCIENCES
INFLUENZA AND OTHER RESPIRATORY VIRUSES
SCIENTIFIC REPORTS
EPIDEMICS
BMC MICROBIOLOGY
INFECTION, GENETICS AND EVOLUTION

Open-source software:

- **Author**, *SynMut*, *Synonymously Mutated Sequences with Different Genomic Signatures*
- **Author**, *GEII*, *Host Gene Expression under Influenza virus infection or Interferon stimulation*
- **Author**, *epitope_diversity*, *Estimating epitope diversity at haplotype level using NGS reads*
- **Author**, *fasta_filter*, *Filtering fasta sequences with threshold of specific bases*
- **Contributor**, *spatPomp*, *Inference for Spatiotemporal Partially Observed Markov Processes*
- **Contributor**, *data.table* (Chinese translation), *High-performance R package*

Professional leadership:

- Founder and sole lecturer of a weekly reading group on Molecular Evolution. Feb. 2025 – Dec. 2025
Personally delivered every session, teaching Ziheng Yang's "Molecular Evolution: The Statistical Approach" to 20 researchers from different regions and disciplines
- Generating and drafting weekly reports to Hong Kong public health authorities. 2020 – 2022
On SARS-CoV-2 genomic epidemiology for timely pandemic responses